Grade 9 Mathematics – Algebraic Fractions

Tutorial – Structured Questions

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Remember: To add or subtract algebraic fractions, first make the denominators equal using the LCM (least common multiple) of the denominators, then add or subtract the numerators. Put brackets around numerators with more than one term before subtracting, and always simplify the final answer.

Instructions: Answer all questions. Show all your working clearly in the space given.

Question 1

Simplify:

(i) $\frac{x}{5} + \frac{2x}{5}$ (ii) $\frac{7a}{9} - \frac{4a}{9}$ (iii) $\frac{x+2}{7} + \frac{2x-3}{7}$ (iv) $\frac{3x+1}{4} - \frac{x-5}{4}$

Question 2

Express each as a single fraction in its simplest form:

(i) $\frac{x}{2} + \frac{x}{3}$ (ii) $\frac{3a}{4} - \frac{a}{6}$ (iii) $\frac{x+1}{3} + \frac{x-2}{4}$

(iv) $\frac{2x-1}{5} - \frac{x-3}{2}$ (v) $\frac{x}{2} + \frac{x}{3} + \frac{x}{4}$

Question 3

Simplify:

(i) $\frac{3}{x} + \frac{5}{x}$ (ii) $\frac{7}{2a} - \frac{3}{2a}$ (iii) $\frac{x+1}{y} + \frac{x-1}{y}$

Question 4

Simplify:

(i) $\frac{2}{x} + \frac{3}{2x}$ (ii) $\frac{1}{a} + \frac{1}{b}$ (iii) $\frac{5}{3x} - \frac{1}{6x}$

(iv) $\frac{3}{4p} + \frac{2}{3p}$ (v) $\frac{2}{x} - \frac{3}{x^2}$ (vi) $\frac{a}{2b} + \frac{b}{3a}$

Question 5

Express as a single fraction:

(i) $\frac{1}{2} + \frac{1}{x}$ (ii) $\frac{x}{3} - \frac{2}{x}$ (iii) $3 + \frac{2}{x}$

Question 6

A student simplified $\frac{x}{2} + \frac{x}{3}$ as follows: $\frac{x}{2} + \frac{x}{3} = \frac{x + x}{2 + 3} = \frac{2x}{5}$

(i) Explain the mistake made by the student.

(ii) Write the correct answer.

Question 7

(i) Simplify $\frac{x + 3}{2} - \frac{x - 1}{3}$.

(ii) Using your answer, find the value of the expression when $x = 1$.

(iii) Check your answer by substituting $x = 1$ directly into the original expression.

Question 8

Kamal has Rs x. He spends one third of it on books and one fourth of it on food.

(i) Write an algebraic expression, as a single fraction, for the total amount he spends.

(ii) Write an expression, as a single fraction, for the amount left.

(iii) If x = 1200, find the amount left.

Question 9

The length of a rectangle is $\frac{x + 1}{2}$ cm and its breadth is $\frac{x - 1}{3}$ cm.

(i) Show that the perimeter of the rectangle is $\frac{5x + 1}{3}$ cm.

(ii) Find the perimeter when x = 7, and check it using the length and breadth.

Question 10

(i) Show that $\frac{x+2}{3} - \frac{x-4}{6} = \frac{x+8}{6}$.

(ii) Hence, solve the equation $\frac{x+2}{3} - \frac{x-4}{6} = 3$.

Question 11

Thinking question: Find the missing term in each box.

(i) $\frac{a}{4} + \frac{\square}{6} = \frac{3a+4b}{12}$ (ii) $\frac{\square}{x} - \frac{1}{3x} = \frac{5}{3x}$